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Physics of the Earth and Planetary Interiors 138 (2003) 183–196
PHYSICS OF THE EARTH AND PLANETARY INTERIORS
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Paleomagnetic poles and paleosecular variation of basalts from Paraná Magmatic Province, Brazil: geomagnetic and geodynamic implications

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Received 10 December 2002; received in revised form 10 July 2003; accepted 23 July 2003

Abstract

A paleomagnetic study was carried out along two sections (altogether 35 lava flows, 300 samples) in the central Paraná Magmatic Province (PMP), Brazil. The two sections, distanced ca. 200 km apart, yield statistically indistinguishable paleomagnetic poles. The combined paleomagnetic pole with coordinates −85.7°N, 197.9°E ($A95 = 2.6°$, $N = 35$) is statistically different from previously published paleomagnetic poles for other sections of PMP. We suggest that this angular difference, as well as differences between previously published poles, is caused by undetected local tectonic rotations not easily identified in the often-poorly exposed lavas of the PMP.

A joint analysis of all published PMP paleomagnetic data indicate that paleosecular variation, estimated as the angular standard deviation (S_F) of the virtual geomagnetic pole distribution, does not support the suggestion of anomalously high secular variation at low latitudes in the 110–195 Ma period [J. Geophys. Res. 96 (1991) 3923]. All S_F estimates are, in fact, in better accordance with latitudinal dependence estimates derived from the curve for the 0–5 Ma period. Moreover, we find that all PMP paleomagnetic poles lie ∼10° away from the pole predicted by an assumed fixed hotspot reconstruction of South America. The PMP paleomagnetic poles, therefore, call for either true polar wander or motion of Indo-Atlantic hotspots.
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Keywords: Paleomagnetism; PSV; Paraná; Brazil

1. Introduction

The period between about 120 and 80 Ma was characterized by globally intense geodynamic processes with widespread magmatic activity, continental breakup, rifting and fast ocean spreading rates (e.g. Larson, 1991; Coffin and Eldholm, 1994; Anderson, 1994). The Early Cretaceous age (∼132–133 Ma) of the Paraná Magmatic Province (PMP) coincides with this period in the Earth's history of major geodynamic changes when several large igneous provinces (LIPs), similar to the PMP, were formed. Coffin and Eldholm (1994) have estimated that giant LIPs account for as much as 50% of the mass and energy expelled from

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doi:10.1016/S0031-9201(03)00158-4

Earth's mantle into the lithosphere in Cretaceous time. In contrast, today's plume-associated volcanism accounts for only about 5–10% (Stein and Hofmann, 1994), suggesting a substantial change in mantle dynamics from the Cretaceous to present. Moreover, widespread evidence indicates that during early to mid Cretaceous the climate was considerably warmer, sea level was significantly higher, and episodes of oceanic anoxia and black shale deposition were more frequent (Larson, 1991; Coffin and Eldholm, 1994; Larson and Erba, 1999). The geodynamo may also have been in an anomalously high-energy state during the Early Cretaceous resulting not only in the Cretaceous normal superchron (Cande and Kent, 1995) but also anomalous high paleosecular variation (PSV; McFadden et al., 1991). It has been suggested that also a fast true polar wander (TPW) event took place at this time (Prévot et al., 2000). Paleomagnetic data from the PMP can help elucidate some of these questions.

The Paraná Magmatic Province (PMP) has been the subject of extensive paleomagnetic studies yielding a large database of paleomagnetic directions (Tamrat and Ernesto, 1999 and references therein). Paleomagnetic data from the PMP are, however, not straightforward as the poor exposure makes it impossible to define paleo-horizontal and it is therefore difficult to assess possible tectonic tilting (e.g. Ernesto et al., 1999) and its influence on the paleomagnetic poles and paleosecular variation estimates. Regardless of this, some units were found almost horizontal as also mentioned in all Paraná paleomagnetic studies (see Ernesto et al., 1990; Tamrat and Ernesto, 1999; Ernesto et al., 1999). In this study, we present new detailed rock-magnetic and paleomagnetic directional data from 35 lavas sampled in the central part of the PMP. We will discuss our results together with previously published ones and their possible connection to true polar wander and paleosecular variation.

2. Geology and sampling

The PMP represents one of the world's largest volumes of Mesozoic continental flood basalts, covering an area about $1.2 \times 10^6\,km^2$, located in southern Brazil (mainly), Uruguay, Paraguay and Argentina. Renne et al. (1992, 1996a,b) showed that the entire PMP erupted in a very narrow age interval (mainly 133–132 Ma) although slightly older ages have been reported (e.g. Turner et al., 1994). Geochemical characteristics, linkage with the present-day Tristan da Cunha hotspot, and the identification of a fossil plume head using seismic tomography indicate that the Paraná-Etendeka LIP was caused by the appearance of a mantle plume. Basalts from the Benue Trough in western Africa range from 138 to 131 Ma (Maluski et al., 1995), and tholeiites from the Kwanza basin (Angola) range from 132 to 126 Ma (Marzoli et al., 1999). This activity in northeastern Brazil and the juxtaposed region of Africa has been named the EQA (Equatorial circum-Atlantic) event (Marzoli et al., 2000). The similarity in age with the Paraná-Etendeka activity might suggest that the EQA activity was also due to the Tristan da Cunha plume. The EQA plume centre must be located in northeastern Brazil near the breakup margin, and is therefore distinct from the Paraná-Etendeka plume of the same age, but probably due to the same thermal disturbance in the deep mantle. These two 138–130 Ma plume centers, located 2500 km apart, are both related to the ensuing separation of South America and Africa during formation of the south Atlantic Ocean.

The PMP is divided into three major parts by the Rio Uruguai and Rio Piquiri tectonic lineaments (Fig. 1A and B), which have existed since Devonian. This division is not only tectonic but also can be traced by geochemical data (Stewart et al., 1996). Tholeitic basalts are the dominant rock type; however, volcanic suites may also include some acid and basic rocks (Ernesto et al., 1990; Tamrat and Ernesto, 1999). Paleomagnetic sampling was carried out in the central part of PMP (Fig. 1B) between both major lineaments. Cores were drilled in the field and oriented in most cases with both magnetic and sun compasses. Two sequences were sampled (35 lava flows, about 300 standard paleomagnetic cores): 156 cores belonging to 18 lava flows were collected at Salto Caixa hydro electrical station and along the Capitan Leonidas – Cascavel road (SC: 25°24′S, 53°34′W); 17 cooling units were sampled along the highway BR153 from Rio Uruguai level to Concordia and Irani (CON: between 27°22′S, 51°59′W and 26°47′S, 51°42′W). It is quite possible that our section SC corresponds to section CI from Tamrat and Ernesto (1999) although no paleomagnetic holes were found at our sampling area.

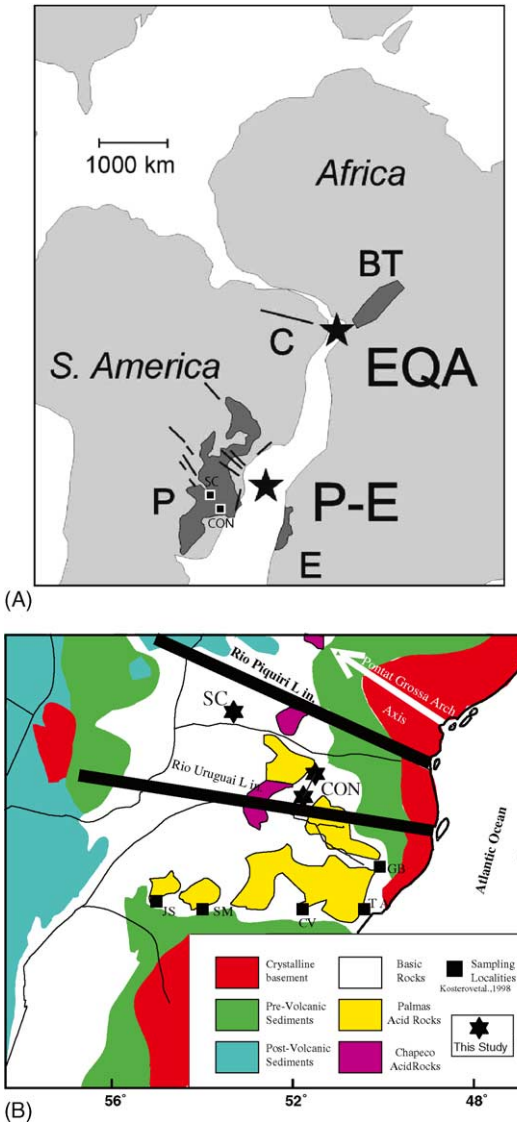


Fig. 1. (A) Ca 135 Ma plume cluster of Paraná-Etendeka event "P-E", the Equatorial Circum-Atlantic event "EQA" comprising the Rio Ceara Mirim dyke, "C", and Benue Trough magmatism, "BT". Stars locate Paraná-Etendeka and the Equatorial Circum-Atlantic plume centers. Sections sampled are indicated by squares (SC and CON). Modified from Ernst and Buchan (2002). (B) Simplified geological map of Paraná flood basalts with indication of paleomagnetic sections, modified from Kosterov et al. (1998).

3. Results

3.1. Microscopy observations

The optical examination of 12 polished sections, from both studied sequences, allows the estimation of opaque minerals concentration subdivided into fractions of titanomagnetite, titanohematite, and titanomaghemite. We followed Buddington and Lindsley (1964) in classifying textures and Haggerty (1976) in resolving oxidation states. The principal results of oxide observations can be summarized as follows:

1. Titanomagnetites are transformed to Ti-poor titanomagnetite.
2. Titanomagnetites show anhedral to subhedral textures with ilmenite lamellae.
3. There is minor (titano)maghemite, and hematite is scarce.

Fig. 2A shows skeletal (titano)magnetite with small hematite veins, curved cracks and the corroded border may suggest the presence of partial maghemitization; a subhedral (titano)magnetite crystal with ilmenite lamellae (sandwich texture) and hematite veins is shown in Fig. 2B; Fig. 2C shows an anhedral (titano)magnetite crystal crossed by large ilmenite lamellae. Finally, Fig. 2D shows two anhedral grains of (titano)magnetite with an ilmenite lamella crossing both crystals.

In summary, the titanomagnetite seems to be the main magnetic carrier in the rocks studied. However, it is not homogeneous with ilmenite intergrowths showing a light gray-brown color, sometimes fractured or with a brecciated texture. The general characteristics of magnetic mineralogy show deuteric oxidation of degree C1 and C2 (Haggerty, 1976) and a possible light hydrothermal alteration. These observations under the microscope suggest that the natural remanent magnetization carried by these grains should be a thermoremanent magnetization, because the described paragenesis typically develops at temperatures higher than 600 °C.

3.2. Rock-magnetic observations

Magnetic characteristics of typical samples are summarized in Figs. 3–5 and are described as follows:

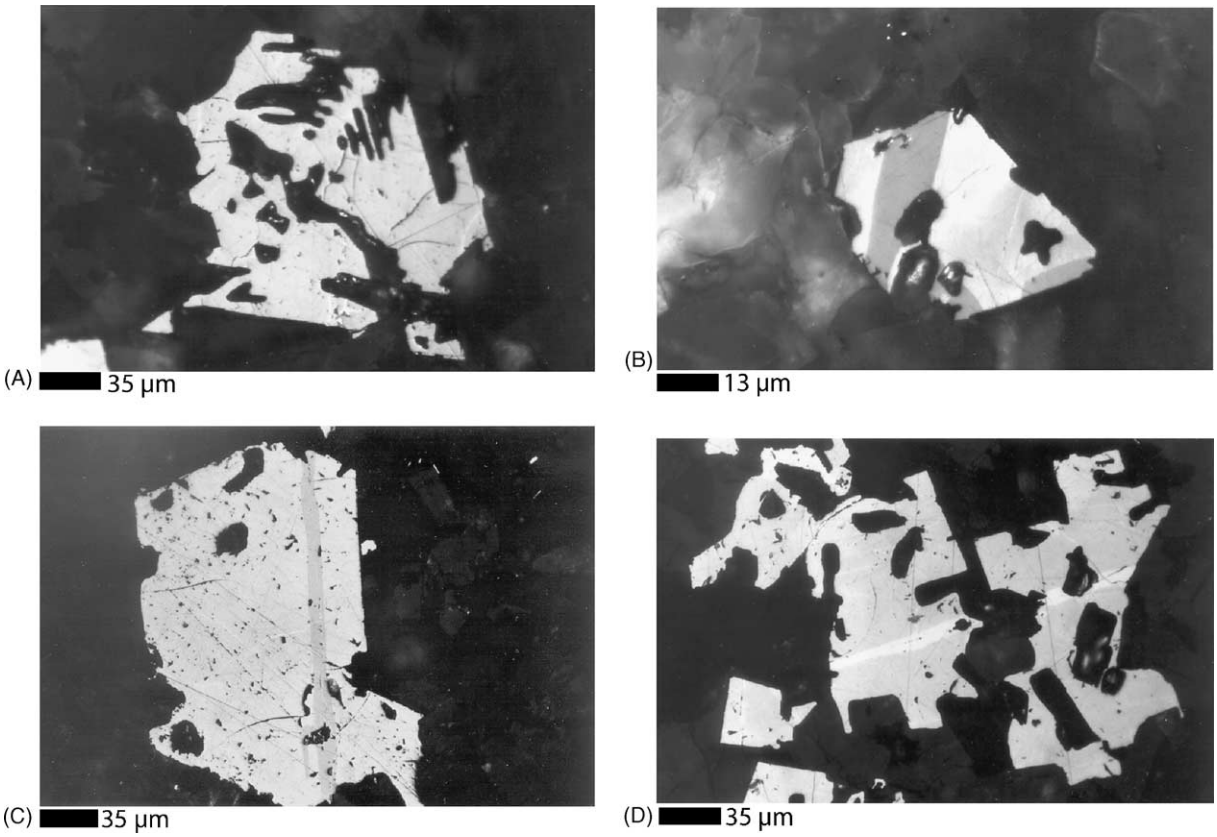


Fig. 2. Photomicrographs representative of the Paraná basaltic rocks.

1. Continuous curves of low-field susceptibility vs. high temperature show, in most cases, the presence of a single ferrimagnetic phase with Curie point compatible with nearly pure magnetite (Fig. 3A and B). However, the cooling and heating curves are occasionally not perfectly reversible (cooling curve notably lowers than heating curve indicating oxidation during the heating in air). Some samples (Fig. 3C and D) curves show an anomalous peak of susceptibility around 350–400 °C on heating that may correspond to maghemite subsequently transformed to magnetite (Readman and O'Reilly, 1972; Özdemir, 1987). Nevertheless, both experimental and theoretical studies (Heider and Dunlop, 1987; Özdemir and Dunlop, 1989; Nishitani and Kono, 1989) show that chemical remagnetization by maghemitization has the same direction as the original TRM. Thus, paleodirections were most possibly unaffected by this alteration and they can be used for determining magnetic polarities and mean paleodirections.
2. Hysteresis measurements at room temperature were performed on all studied samples using the AGFM 'Micromag' apparatus of the paleomagnetic laboratory at Mexico City in fields up to 1.4 T (Fig. 4). The curves are symmetrical in all cases (Fig. 4A and B). Near the origin, potbellied and wasp-waisted behavior (Tauxe et al., 1996) was not detected, which probably reflects very restricted ranges of the opaque mineral coercivities. The only exception is site SC03 (Fig. 4C), which probably reflects the presence of ferrimagnetic phases with different coercivities. Most probably this behavior is due to the presence of dominantly single domain and superparamagnetic grains (Goguitchaichvili and Prévot, 2000). For other samples (Fig. 4A

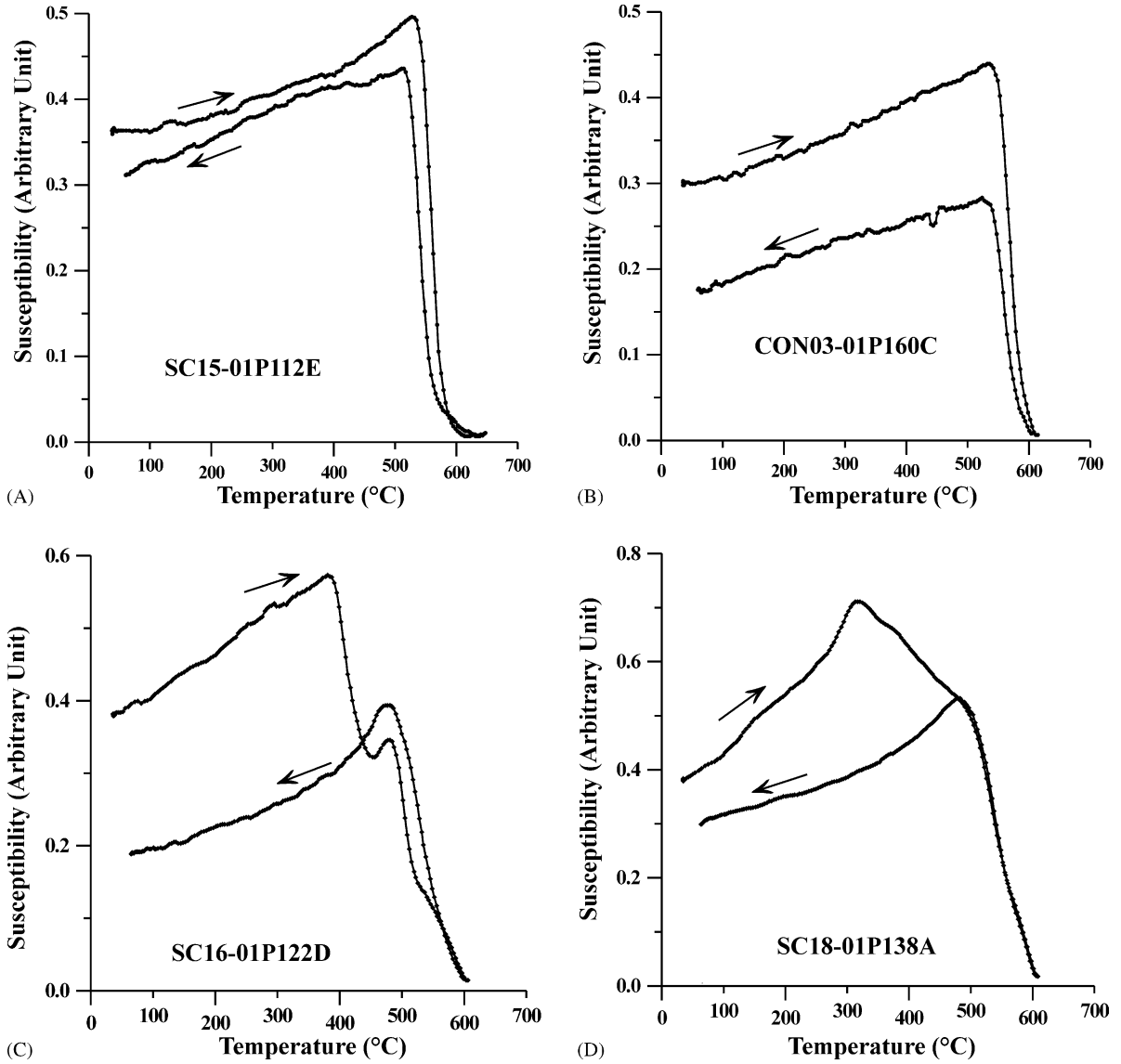


Fig. 3. Susceptibility versus temperature curves of representative samples. The arrows indicate the heating and cooling curves.

and B), 'small' pseudo-single-domain grains seem to be responsible for remanence, judging from hysteresis parameters values (Day et al., 1977; Fig. 5). This probably indicates a mixture of multidomain and a significant amount of single-domain (SD) grains. Let us note that if some superparamagnetic fraction also exists in these samples, the measured coercive force and saturation magnetization are somewhat lower or larger, respectively, than those ferrimagnetic fractions alone.

3.3. Paleodirectional results

Both standard thermal and alternating field demagnetization procedures were used to determine paleomagnetic directions. Characteristic remanent

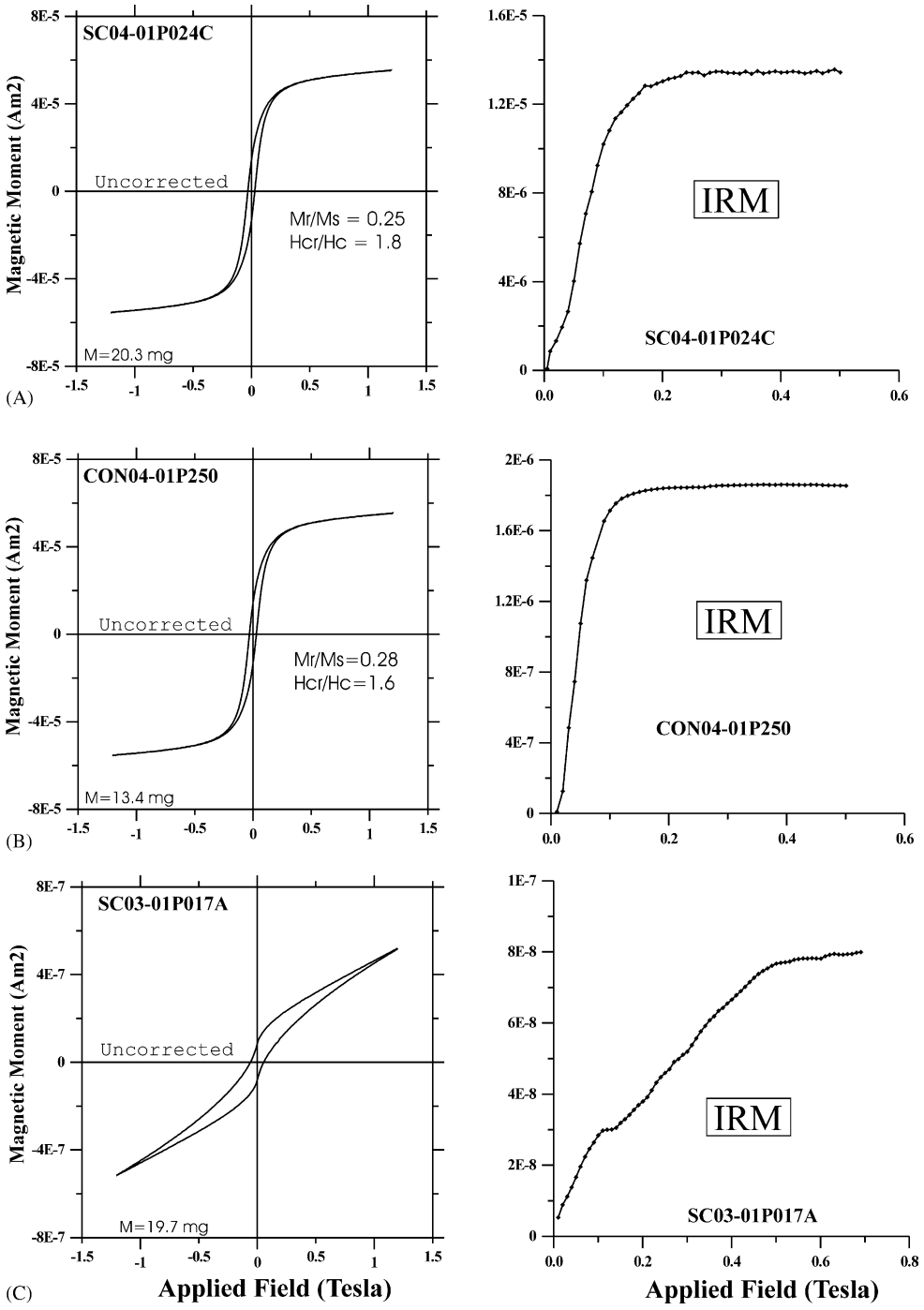


Fig. 4. Examples of hysteresis loops (uncorrected) of small chip samples (see also text) and corresponding isothermal remanent magnetization (IRM) acquisition curves.

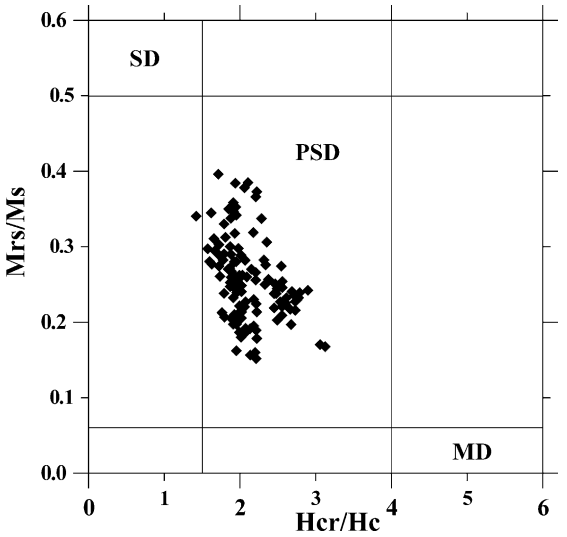


Fig. 5. Magnetic hysteresis parameter ratios plotted in a Day diagram (Day et al., 1977). Data for the basalt flows fall into the pseudo-single-domain (PSD) range.

magnetization (ChRM) directions were isolated by means of principal component analysis (Kirschvink, 1980), and flow mean directions were calculated in general on 6–9 samples per flow.

Most of the samples studied carry a single, stable component of magnetization, observed upon both alternating field (Fig. 6A, E, and F) and thermal (Fig. 6B–D) treatment. A commonly minor secondary component, probably of viscous origin, is present but was easily removed applying 20 mT or 250 °C. In one case (site SC17), ChRM direction was obtained after applying 70 mT. The median destructive fields range mostly in the 40–60 mT interval, suggesting the existence of small pseudo-single (mainly) to single domain magnetic grains as remanence carriers (Dunlop and Özdemir, 1997). Higher coercivity was observed for site SC10. The larger part of remanent magnetization in most cases was removed at temperatures between 530 and 570 °C, which suggest that low-Ti titanomagnetites are responsible for the magnetization.

Paleomagnetic directions are well grouped (Table 1 and Fig. 7) at the within-site (flow) level, with Fisher precision parameter k generally ranging from 21 to 389 and α_{95} ranging from 12.6 to 2.8°. A bootstrap statistical test (Fig. 8) shows that the two studied sections share a common mean. All sites yielded normal magnetic polarity. The mean paleodirection obtained for section SC is $I = -42.4$, $D = 7.0°$, $k = 108$, $\alpha_{95} = 3.3°$, $N = 18$; and for section CON $I = -42.4$, $D = 1.6°$, $k = 93$, $\alpha_{95} = 3.3°$, $N = 17$. The mean overall paleodirection obtained in this study is $I = -42.4$, $D = 4.4°$, $k = 97$, $\alpha_{95} = 2.5°$, $N = 35$ (Table 1).

4. Discussion

The PMP has been extensively studied with altogether 672 paleomagnetic sampling sites recording the direction of the paleomagnetic field (Table 2). This extensive dataset allows us to discuss paleosecular variation and test differences between paleomagnetic and hotspot reconstructions for South America.

4.1. Paleosecular variation

The classical formula $S_F^2 = S_T^2 - S_W^2/n$ was used for estimating paleosecular variation in this study. Here, S_T is total angular dispersion $S_T = [(1/N-1)\sum_{i=1}^{N}\delta i^2]^{1/2}$ (Cox, 1969), N is the number of sites used in the calculation, δi is the angular distance of the i-th VGP (virtual geomagnetic pole) from the axial dipole. S_W represents the within site dispersion, n is the average number of samples per unit. The paleosecular variation (SF) of the virtual geomagnetic pole (VGP) distribution is determined with the implicit assumption that the distribution is Fisherian (Fisher, 1953). In order to test whether our 35 VGPs are Fisher distributed we produce quantile–quantile plots for the exponential colatitudes and the uniform longitude (Fig. 9). Both plots are seen to be more or less linear, and with M_u and M_e values lower than 1.207 and 1.094, respectively, and a Fisher distribution hypothesis cannot be rejected at the 95 percent confidence level (Tauxe, 1998). We note that the pole of Ernesto et al. (1999) (study 4 in Table 2) has a slightly non-Fisherian distribution of VGPs (as pointed out by the authors) suggesting either non-removal of secondary overprints or differential tectonic disturbance of their sampling sites.

The paleosecular variation estimate for this study is shown in Fig. 10A together with the predicted VGP scatter for 110–195 Ma and 0–5 Ma (McFadden et al.,

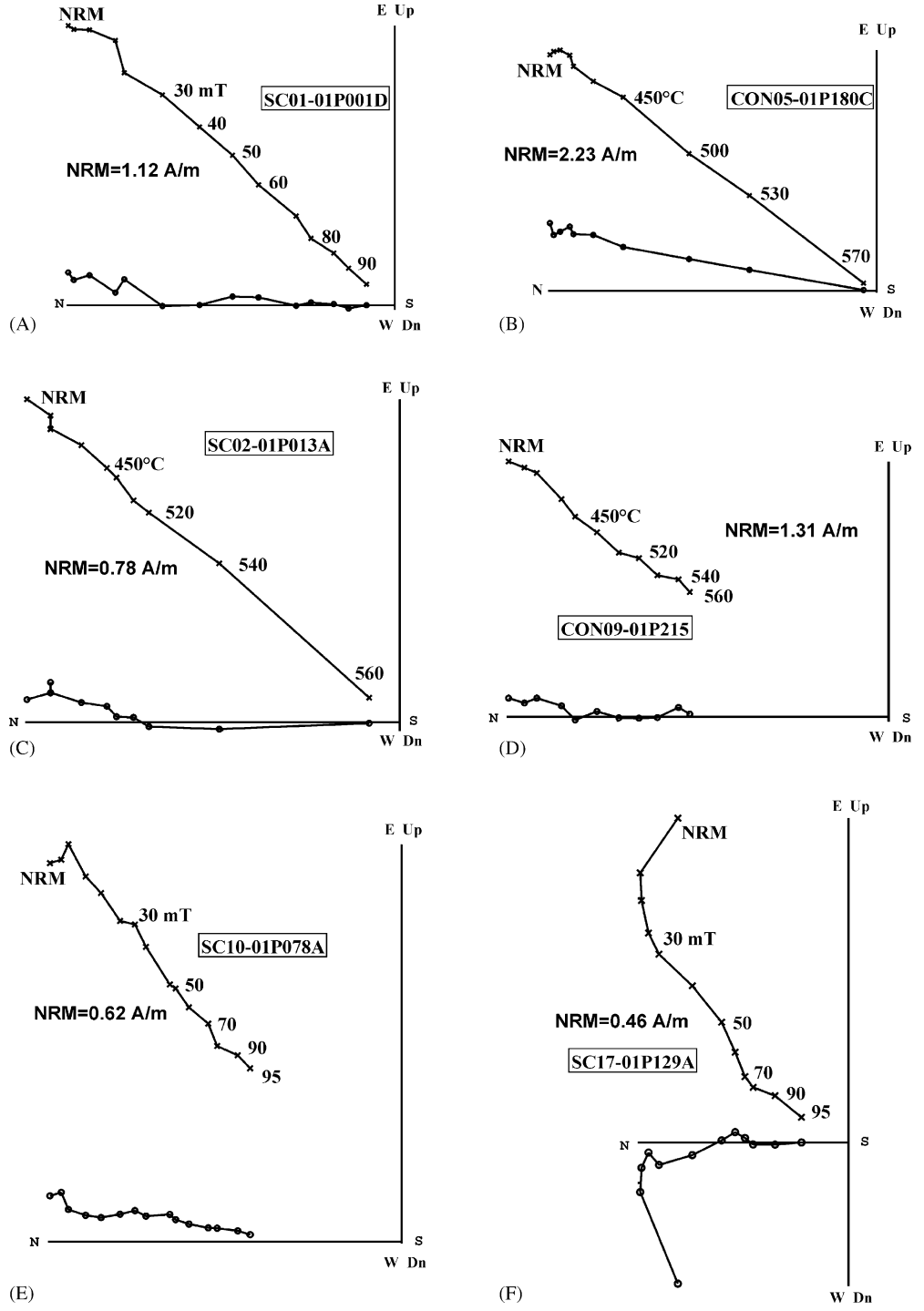


Fig. 6. Orthogonal vector plots of stepwise alternating field and thermal demagnetization (stratigraphic coordinates). The numbers refer to peak alternating fields in mT or temperatures in °C. (○) projections into the horizontal plane, (×) projections into the vertical plane.

Table 1
Paleomagnetic results of two studied sections from Paraná

Site	Site Lat	Site Long	*n*/*N*	Inc (°)	Dec (°)	*k*	a_{95} (°)	Plat (°N)	Plong (°E)
SC01	−25.46	306.42	9/9	−44.2	6.1	339	3.1	−84.5	222.7
SC02	−25.46	306.42	7/7	−39.3	11.8	116	7.1	−78.7	202.1
SC03	−25.46	306.42	6/6	−50.2	359.9	82	6.5	−84.5	307.3
SC04	−25.46	306.42	8/9	−41.4	7.7	169	4.3	−82.8	204.3
SC05	−25.46	306.42	8/8	−40.6	10.4	154	4.6	−80.3	205.2
SC06	−25.46	306.42	7/8	−46.3	11.4	98	5.6	−79.6	230.9
SC07	−25.46	306.42	8/9	−43.1	354.1	78	6.2	−84.7	39.3
SC08	−25.46	306.42	6/6	−39.9	3.5	232	3.6	−85.7	175.8
SC09	−25.43	306.42	7/8	−45.9	15.6	193	3.9	−75.9	227.5
SC10	−25.40	306.42	9/9	−42.7	4.1	219	3.7	−86.2	207.7
SC11	−25.38	306.42	4/5	−48.9	14.6	43	8.8	−76.3	238.7
SC12	−25.38	306.42	7/7	−45.2	8.6	102	5.7	−82.2	228.2
SC13	−25.38	306.42	7/8	−40.8	355.4	67	7.4	−85.3	61.4
SC14	−25.38	306.42	7/8	−46.1	22.3	136	4.8	−69.9	227.3
SC15	−25.38	306.42	7/7	−30.8	8.7	82	6.3	−78	170.8
SC16	−25.38	306.42	6/8	−40.1	12.3	55	7.7	−78.5	206.2
SC17	−25.31	306.42	7/9	−28.6	355.6	204	3.9	−79.1	103.3
SC18	−25.21	306.40	7/7	−44.7	7.9	138	4.8	−82.8	227.1
CON01	−27.37	308.01	8/9	−53.7	9.9	56	7.4	−79.1	259.4
CON02	−27.37	308.00	7/9	−50.1	5.5	31	10.1	−84.1	255.2
CON03	−27.34	308.02	6/9	−49.1	7.5	116	5.2	−82.9	241.7
CON04	−27.32	308.02	8/8	−40.8	358.8	76	6.6	−85.9	112.4
CON05	−27.31	308.01	7/8	−30.9	10.3	389	2.8	−75.7	172
CON06	−27.30	308.02	7/8	−42.3	8.1	89	5.9	−82.2	198.5
CON07	−27.24	308.05	8/8	−43.4	9.2	138	4.7	−81.5	207.2
CON08	−27.22	308.05	7/7	−42.7	3.5	44	8.5	−86	181.1
CON09	−27.21	308.05	8/8	−31.2	2.8	51	7.8	−79.3	142.7
CON10	−27.08	308.10	6/8	−35.2	5.5	27	10.7	−80.8	162.6
CON11	−27.07	308.10	7/7	−39.1	5.8	37	9.2	−82.7	175.9
CON12	−27.05	308.12	8/8	−45.4	4.2	70	6.6	−86.3	216.6
CON13	−27.03	308.12	7/8	−44.9	349.7	73	6.5	−80.8	39
CON14	−27.03	308.12	7/7	−43.9	345.8	47	8.2	−77.2	40.8
CON15	−26.83	308.27	6/8	−37.8	347.6	21	12.6	−77.4	62
CON16	−26.80	308.30	8/8	−44.5	354.6	174	4.2	−85.1	44.5
CON17	−26.77	308.33	7/8	−41.6	359.6	144	4.6	−87.1	120.9
Mean			249	−42.4	4.4	97	2.5	−85.7	197.9

Site Lat, Site Long are the geographic co-ordinates of each lava flow; *n*/*N* is the number of used/treated samples; Inc, Dec are inclination and declination of the site-mean ChRM; *k* and a_{95} are the precision parameter and the confidence cone of the Fisher statistics (Fisher, 1953); Plat, Plong are the latitude and longitude of the virtual geomagnetic pole.

1991). The vast difference in paleosecular variation for low latitudes between the 0–5 Ma and 110–195 Ma curve has been interpreted by McFadden et al. (1991) to indicate an anomalous state of the geodynamo in the Cretaceous. The VGP scatter determined for the 35 lava flows in this study is slightly smaller than the values calculated using the data in other studies of PMP sequences (Fig. 10B). This difference may be due to sampling bias (having sampled the same time units) or too short interval to average the paleosecular variation as extrusion rates in continental flood basalt provinces have been shown to be high, and relatively thick volcanic sequences may represent short time intervals (e.g. Renne et al., 1992). The stratigraphy for our sampling sites is difficult to determine due to the complexity of Paraná terrain. The relative stratigraphy between two studied sections (SC and CON some 200 km apart) cannot be estimated because of

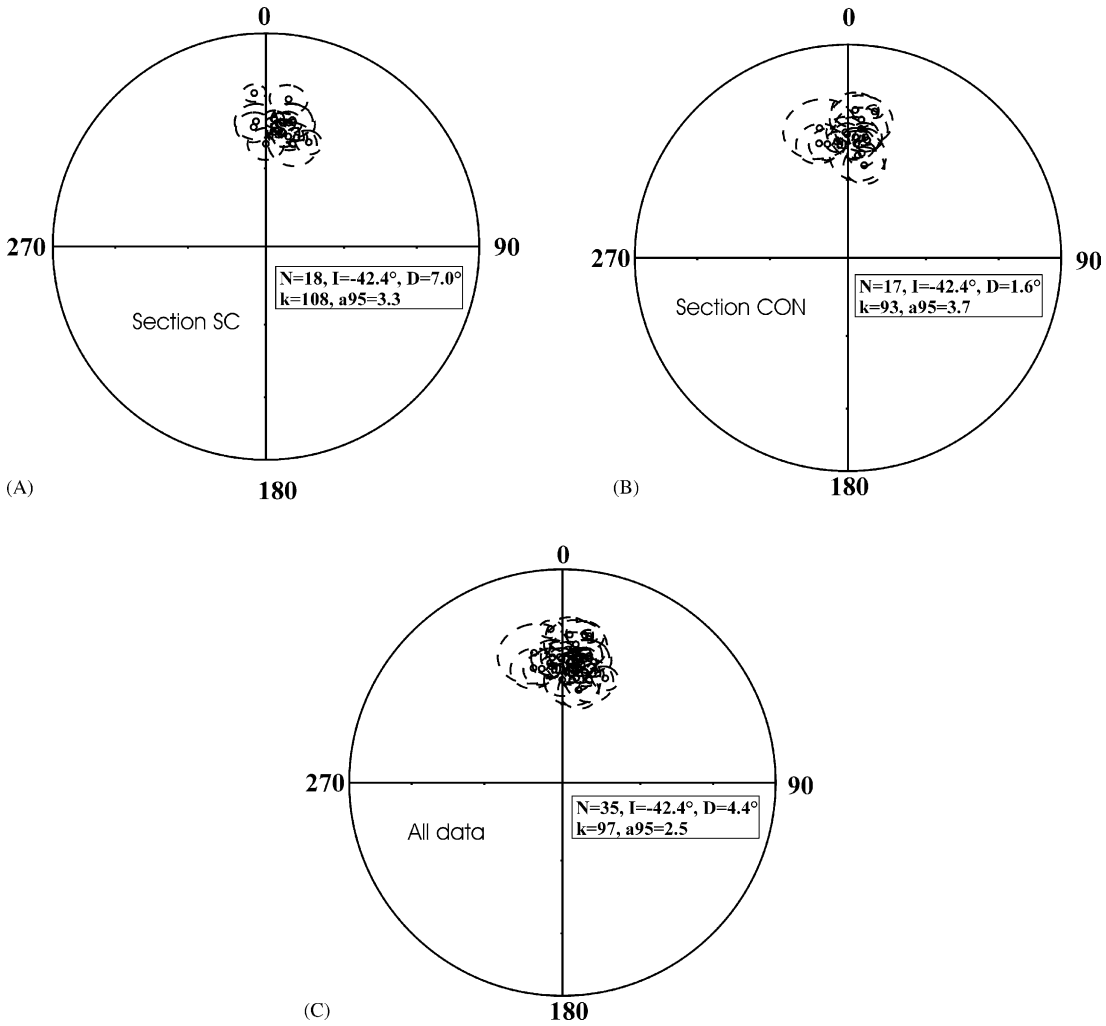


Fig. 7. Equal area projections of the flow mean characteristic paleodirections for studied sections.

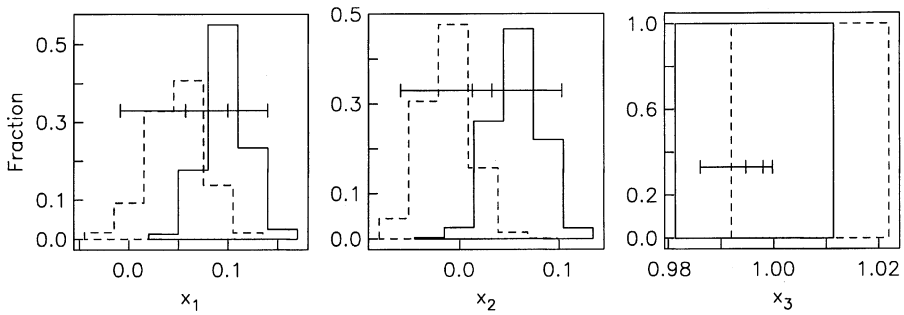


Fig. 8. Histograms of x_1, x_2, x_3 components of the bootstrapped means from CON and SC sections' data sets. Because the confidence intervals for all components overlap, the two sets of bootstrapped means cannot be distinguished at the 95% level of confidence; they pass the bootstrap test for a common mean (Tauxe, 1998).

Table 2
Selected paleomagnetic poles for 120–140 Ma

Sampling location	Age (Ma)	N	Lat. (°N)	Long. (°E)	a_{95} (°)	ASD (°)	Reference
Central PMP	~132	35	−85.7	197.9	2.6	8.5 + 1.7/−1.2	This study
Southern PMP	~133	197	−84.0	106.2	1.5	12.1 + 0.9/−0.8	Raposo and Ernesto (1995)
Central PMP	~132	103	−84.1	64.4	2.3	13.1 + 1.4/−1.2	Raposo and Ernesto (1995)
Northern PMP	~132	92	−83.0	71.4	2.4	13.0 + 1.5/−1.2	Ernesto et al. (1999)
Ponta Grossa	~130	115	−82.4	30.3	2.0	12.2 + 1.2/−1.0	Raposo and Ernesto (1995)
Alkaline Province Paraguay	127–130	75	−85.4	62.3	3.1	15.0 + 1.9/−1.5	Ernesto et al. (1996)
Cordoba Province Argentina	~125	55	−86.0	75.9	3.3	–	Geuna and Vizan (1998)
Kaoko basalts[a]	~132	–	−84.7	103.9	3.2	–	Gidskehaug et al. (1975)
Fixed hotspot[b]	~130	–	−76.7	116.4	–	–	Muller et al. (1993)

N is the number of sites used for determination; Lat., Long. are the latitude and longitude of the paleomagnetic pole; a_{95} is the confidence cone of the Fisher statistics (Fisher, 1953); S_F is the estimated angular standard deviation. For the studies other than the present, S_F value was determined using formula (6.20) in Butler (1992) with 95% confidence limits calculated after Cox (1969).

[a] Rotated to South America from Africa (Geuna et al., 2000).

[b] Location of South American paleomagnetic pole predicted from fixed hotspot rotations (Muller et al., 1993).

the presence of several fault systems and lineaments. The sites in Table 1 are shown in apparent stratigraphic order with consecutive flows having higher altitudes. However, due to the complex topography their true stratigraphic position cannot be ascertained. Ignoring the relative small difference between different PMP studies it is noteworthy that all five PMP paleosecular variation estimates (Fig. 10B) for the PMP lava sequences are fairly consistent and considerably smaller than that predicted for the 110–195 Ma curve. These observations are not only of importance for discussions of the Early Cretaceous geodynamo. Since the paleosecular variation model of McFadden et al. (1991) is often used to determine whether PSV have been averaged out, it is important to note that the model may not be representative for the Early Cretaceous field. Keeping the limitations into account, we tentatively suggest for future study that the relatively well-defined latitudinal dependence of angular dispersion documented for the last 5 Ma paleosecular variation could also be representative for Early Cretaceous time.

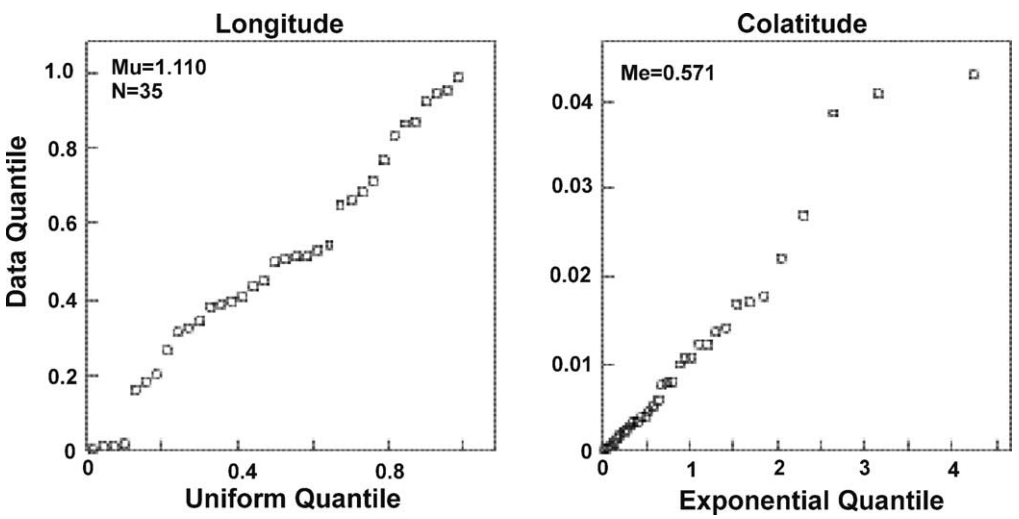


Fig. 9. Quantile–quantile plots assessing goodness-of-fit of the Fisher distribution to VGPs. Value M_u is lower than 1.207 and M_e is lower than 1.094, hence the hypothesis of Fisher distribution cannot be rejected.

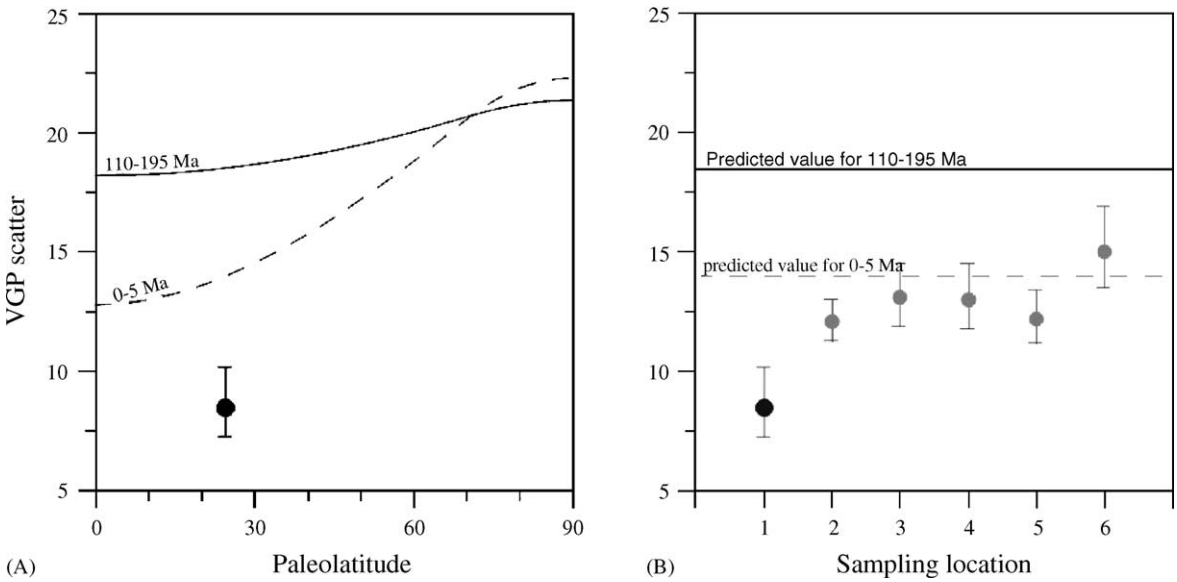


Fig. 10. (A) Paleosecular variation for lavas for 110–195 Ma (solid line) and 0–5 Ma (stippled line) (after McFadden et al., 1991). The angular standard deviation (S_F) value for this study ($8.5 + 1.7/-1.2$) is shown as a closed circle. (B) Comparison of S_F value of this study with S_F values of similar age (see Table 2 for further details). Numbering of sampling locations is from top to bottom in Table 2.

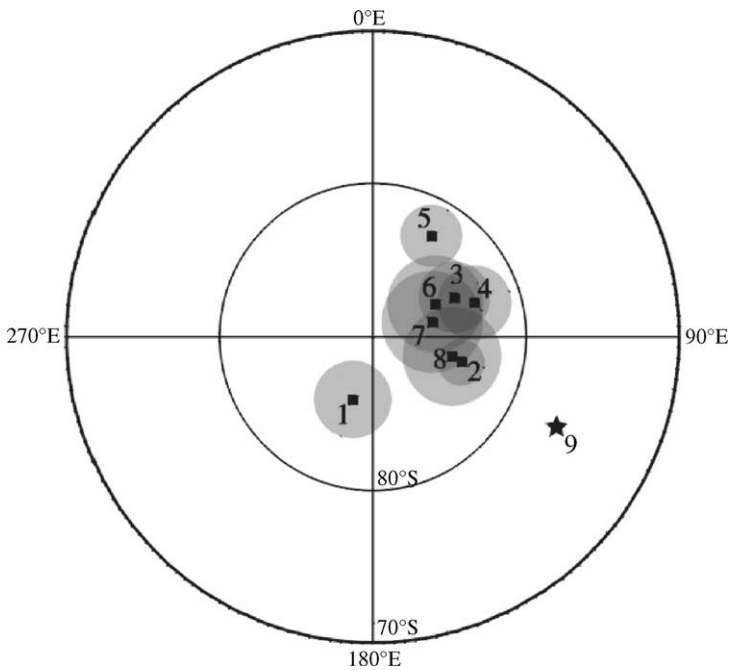


Fig. 11. The paleomagnetic pole for this study together with other paleomagnetic poles of the same age listed in Table 2. Numbering of the poles is from top to bottom in Table 2. Fisher 95 percent confidence limits are also shown. Star is for the South American paleomagnetic pole derived from fixed hotspot rotations (Muller et al., 1993).

4.2. Paleomagnetic poles

The mean paleomagnetic pole of this study (−85.7°N, 197.9°E) is shown in Fig. 11 together with previously published PMP poles and a paleomagnetic pole from Africa with similar age rotated to the South America reference frame (Table 2). The fact that our pole is statistically distinct from all previously published poles is attributed to local tectonic rotations. At all our sampling sites the exposure was too poor to allow the paleo-horizontal to be defined. We note that all PMP poles, not only of this study, are somewhat different, indicating unrecognized tectonic disturbances (see discussion in Ernesto et al., 1999).

Also shown in Fig. 11 is the paleomagnetic pole predicted by the hotspot reconstructions suggested by Muller et al. (1993). The fact that all PMP paleomagnetic poles are lying ca. 10° away from the hotspot predicted pole is in accordance with the findings of Besse and Courtillot (1991) and Prévot et al. (2000) clearly suggesting a difference between paleomagnetic and hotspot reconstructions. It is not clear whether this difference reflects true polar wander (Prévot et al., 2000; Camps et al., 2002) or non-fixity of Indo-Atlantic hotspots (Tarduno and Gee, 1995; Tarduno and Smirnov, 2001, 2002; Riisager et al., 2003).

5. Conclusions

The paleosecular variation estimates for PMP lava sequences are lower than suggested by the model of McFadden et al. (1991) for the interval 110–195 Ma. The PMP estimate is in better accordance with the S_F latitudinal dependence curve documented for the recent 0–5 Ma interval (McElhinny and McFadden, 1997). Possible connections between surface and deep interior processes have been suggested from the temporal coincidence of the constant geomagnetic polarity state during the Cretaceous normal superchron and the intense global geodynamic events occurring during the Cretaceous (e.g. Larson, 1991). A possible mechanism involves superplumes arising from the core-mantle boundary, linking deep events across the mantle and lithosphere. The paleodirectional data derived for the PMP sequences shows no paleosecular variation evidence that the geodynamo was in an unusual state during the Early Cretaceous.

The paleosecular variation model of McFadden et al. (1991) is not only of importance for geodynamo implications, it has also often used to evaluate whether PSV have been averaged out. We tentatively suggest, for further investigation, the question of weather the relatively well-defined 0–5 Ma paleosecular variation curve is also representative for the Early Cretaceous.

The PMP paleomagnetic poles are significantly different from the pole position suggested by hotspot reconstructions. It is unclear whether this difference represents true polar wander or hotspot motion.

Acknowledgements

The authors acknowledge the financial support given by DGAPA-UNAM grant IN100100 and CONACYT project J32727-T. We gratefully acknowledge the help of M.L. Rivas who got the microphotographs of oxides minerals and to G. Aguilar for the preparation of polished sections (Lab. Sedimentologia y Diagenesis, IMP).

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